

Yuanlong Zhou

 ryanbowz@outlook.com |  (+1) 979 344 4239 |  <https://ryanbowz.github.io/> |   ryanbowz

Experience

TikTok US | AI Software Engineer

Jan 2025 – Present

- **Architected a Multi-Agent Game Generation platform** (Spec, Design, Code, Asset, Review) to transform user intent into interactive WebGL games. Powered a “playable TikTok” experience across an ecosystem of **700M+ posts** and **55M+ creators**.
- **Built a multimodal AIGC model evaluation and data-curation framework** for game assets (texture, sprite, UI, animation). Designed structured **query matrices, benchmarks, failure taxonomies, condition labels, and positive/negative examples** to systematically measure Seedream-4.5 capability boundaries across viewpoint/orientation, seamless texture, style consistency, and asset usability. Identified recurring failure patterns through quantitative analysis and formulated **post-training hypotheses**. Translated findings into **skill-level conditioning and generation constraints**, improving asset-quality score from **58.4 to 78.2**.
- Built a **Code Agent execution runtime** that harnesses LLM inference with deterministic and sandbox execution. Developed **portable checkpoint recovery** by persisting LangGraph states at super-step boundaries, enabling internal retries and partial regeneration. Task completion improved from **63.2% to 91.5%**, latency reduced by **33%** (4m35s → 3m05s), token consumption dropped **~50%**.
- **Constructed an agentic self-improvement loop with automated evaluation** using VLM scoring, tracing, failure analysis, validation and regression testing. Increased benchmark score from **48.2 to 89.5**, compressing iteration cycles from **1 week to 8 hours**.
- Engineered **scalable infrastructure** for LLM and multimodal AIGC workloads, implementing **queue scheduling, rate limiting, model routing, MQ/FaaS async workers, and Redis caching** under QPS constraints. Re-architected **distributed Go/RPC services** serving **8M+ participants**, decoupling request ingestion from compute bottlenecks, increasing concurrency throughput by **200%**.

Amazon | Applied Scientist Intern

May 2024 – Aug 2024

- Architected a **multimodal AIGC/CV pipeline** for marketing-asset generation: **diffusion generation, SAM segmentation, style/color transfer, semantic embeddings, and layout composition** to generate and refine banners, icons, and product creatives.
- **Designed asset-quality evaluation metrics and optimization algorithms** for multimodal AIGC outputs. Analyzed and characterized failure modes, developed targeted algorithms for **quality filtering/reranking/regeneration, text tokenization and line wrapping, and block-based layout optimization**, improving production asset quality pass rate from **40% to 95%**.
- Productionized the asset generation system on **AWS** to support **200+ campaigns** for **1M+ assets**, reducing average campaign launch time **from 2 weeks to 3 days**. Built scalable infrastructure and CI/CD workflows on S3, EC2, Cloud9, CodePipeline, and SNS.

Google Summer of Code (Rocket.Chat) | Open-Source Developer

May 2024 – Aug 2024

- Built and deployed an open-source **LLM-powered coding assistant** serving **50K+** global users, deployed Mistral and Llama models to support conversational **code generation and editing** within Rocket.Chat developer community workflows.
- **Fine-tuned Mistral using LoRA-based supervised fine-tuning (SFT)** on public code instruction datasets covering **code synthesis, formatting, and bug fixing**, adapting to community coding-assistant workflows and improving unit-test pass rate by **10%**.
- Constructed **multi-turn context and session-state management** with dynamic model routing based on context limits, latency, and inference cost. Integrated GitHub repository actions through **REST APIs** and **OAuth2**.

Texas A&M University | Research Assistant

Feb 2024 – Dec 2024

- Developed a **real-time 3D reconstruction and spatial-computing system** for interactive XR, spanning **plane segmentation, point-cloud clustering, scene graphs, and furniture detection/segmentation/tracking** to reconstruct indoor environments from visual inputs. Published **AURORA: Automated Unleash of 3D Room Outlines for VR Applications** (ACM SIGGRAPH VRCAI) with patents.
- Built real-time **neural 3D rendering and view-synthesis systems** using Gaussian Splatting, WebGL/GLSL, and CUDA. Optimized character animation using GPU rendering, skeleton blending and dual-quaternion skinning, enhancing runtime FPS by **20%**.

Dots Rhythm Technology | Game Developer

Apr 2021 – Apr 2023

- Developed and released **5+ Unity3D games on Steam**, owning full lifecycle from gameplay design, development to release.
- Built real-time graphics and simulation systems across **shaders, lighting, fog, physics, IK, collision, and AI navigation**. Integrated with multithreading, profiling, LOD and visibility culling, reducing P99 frame time by **65%** and draw calls by **50%**.

Languages: Go, Python, TypeScript/Node.js, C/C++, Java, C#, JavaScript, Ruby, Lua, SQL

AI/ML: Multimodal AIGC, LLM Post-Training, SFT / LoRA, Model Evaluation, Diffusion, Computer Vision, SAM, RAG, Agentic Systems

Graphics / 3D: CUDA, GPU Computing, WebGL/OpenGL, GLSL, 3D Reconstruction, Gaussian Splatting, SLAM, View Synthesis, Unity3D

Systems / Infrastructure: Distributed Systems, RPC, Caching, MQ/Kafka, FaaS, Redis, Docker Sandbox, Kubernetes, CI/CD, AWS

Education

Texas A&M University

Master of Computer Science

GPA: 3.8 / 4.0

Aug 2023 – Dec 2024

Sichuan University

Bachelor of Computer Science (minor: Finance)

GPA: 3.8 / 4.0

Sep 2019 – July 2023